

		downwards, the charge is positive.
25	D	Let the velocity of the rod down the slope be v. e.m.f. induced = $B L v \cos \theta$ current induced = e.m.f. induced / R Magnetic force = $B I L$ to the left When terminal velocity is achieved, Net F along slope = 0 Resolving forces along the slope, $mg \sin \theta = \text{magnetic force} \cos \theta$ $= B \times \frac{BLv \cos \theta}{R} \times L \cos \theta$ $v = \frac{mgR \tan \theta}{B^2 L^2 \cos \theta}$

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B

Note that as the value of V is a straight line with a constant gradient, the square of